Patrick Foulis
CS455 Computer Networks
Lab 1 Environment Setup
August 27, 2026

Platform Statement

I am using macOS Tahoe version 26.6.2, and I did not need to install WSL as I am not on a Windows machine.

Full output of *check-setup.sh*

(venv) patrickfoulis@Patricks-MacBook-Air-4 netlab % cd ~/netlab
chmod +x check-setup.sh
./check-setup.sh

```
==========================================================================
 Computer Networks -- Lab 1 environment check
==========================================================================
Run at: Tue Aug 25 12:00:00 EDT 2026
== Platform
  uname: Darwin 25.6.0 arm64
  [ OK ] macOS 26.6.2
== C / C++ toolchain
  [ OK ] gcc -- Apple clang version 21.0.0 (clang-2100.1.1.101)
  [ OK ] g++ -- Apple clang version 21.0.0 (clang-2100.1.1.101)
  [ OK ] make -- GNU Make 3.81
  [ OK ] lldb -- lldb-2100.0.17.203 (macOS debugger; fine)
== POSIX socket headers (required by Beej's Guide)
  [ OK ] All required socket headers present and compilable
== Socket program smoke test
  [ OK ] smoke.c compiled cleanly
  [ OK ] smoke ran -- resolved example.com -> 172.66.147.243, socket fd = 6
== Python 3
  [ OK ] python3 -- Python 3.14.7
  [ OK ] Python version is 3.8 or newer
  [ OK ] pip -- pip 26.2.1
  [ OK ] venv module available
  [ OK ] Virtual environment ACTIVE: /Users/patrickfoulis/netlab/venv
```

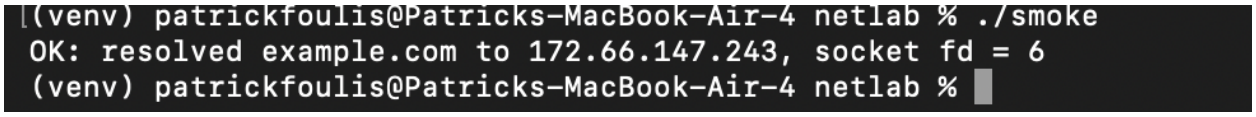
```
  [ OK ] scapy importable inside venv
== Capture tools
  [ OK ] tcpdump -- tcpdump version 4.99.1 -- Apple version 158
  [WARN] tcpdump had permission to open 'lo0' but saw no traffic in 8s. Permissions look fine;
nothing was talking on loopback.
  [ OK ] tshark -- TShark (Wireshark) 4.6.8 (v4.6.8-0-ge677bf052328).
  [ OK ] BPF capture devices are readable (ChmodBPF looks installed)
== Supporting tools
  [ OK ] git -- git version 2.50.1 (Apple Git-155)
  [ OK ] curl -- curl 8.7.1 (x86_64-apple-darwin25.0) libcurl/8.7.1 (SecureTransport)
LibreSSL/3.3.6 zlib/1.2.12 nghttp2/1.68.1
  [ OK ] man pages for syscalls installed (man 2 socket works)
================================================================================

  Passed: 20   Warnings: 1   Failures: 0
  No failures, but read the warnings above.
  Some of them (capture permissions, venv) will matter in Lab 2.
================================================================================
```

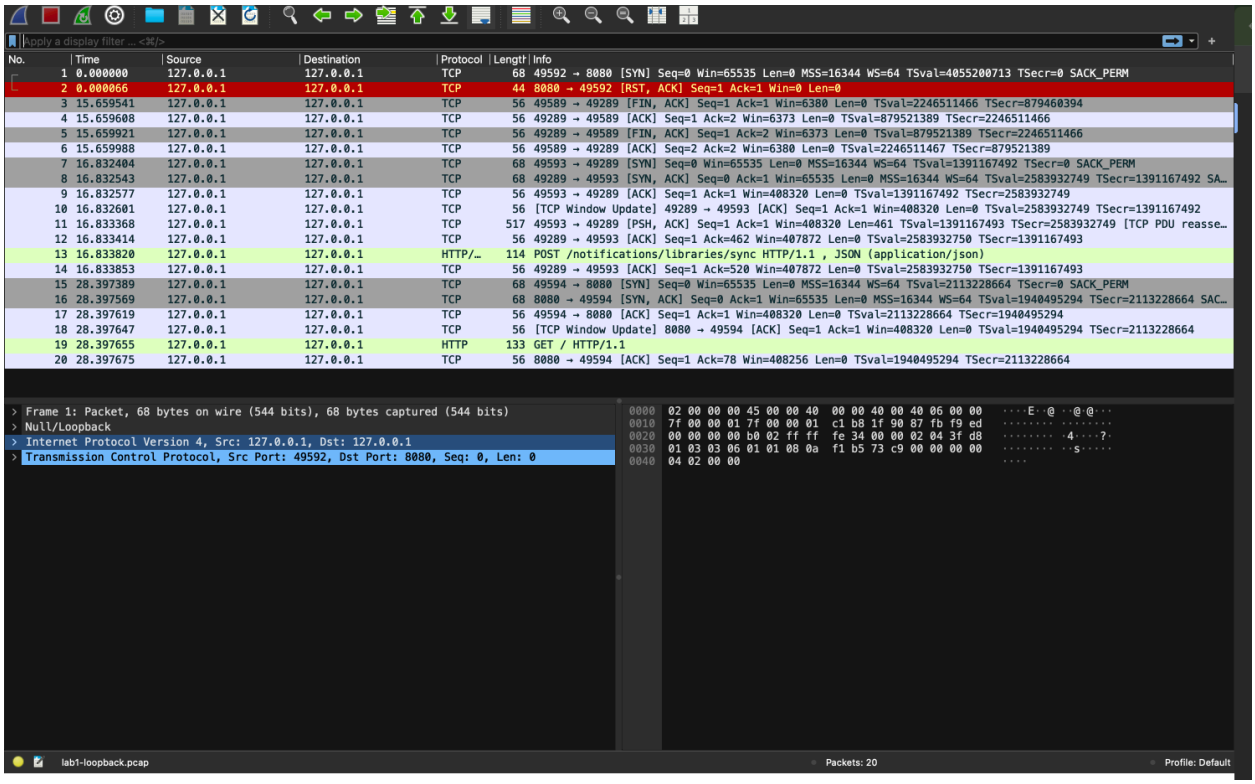
I do have a side note on the Warining. The initial command in the assignment did not work. I used Gemini AI to interpret the error and propose a solution. It came up with this: “On macOS, the loopback interface is named lo0, not lo (which is the standard naming convention on Linux). Run the command using lo0”

I did this and it ran perfectly fine. If this matters in the future I will adjust but I dont think it will.

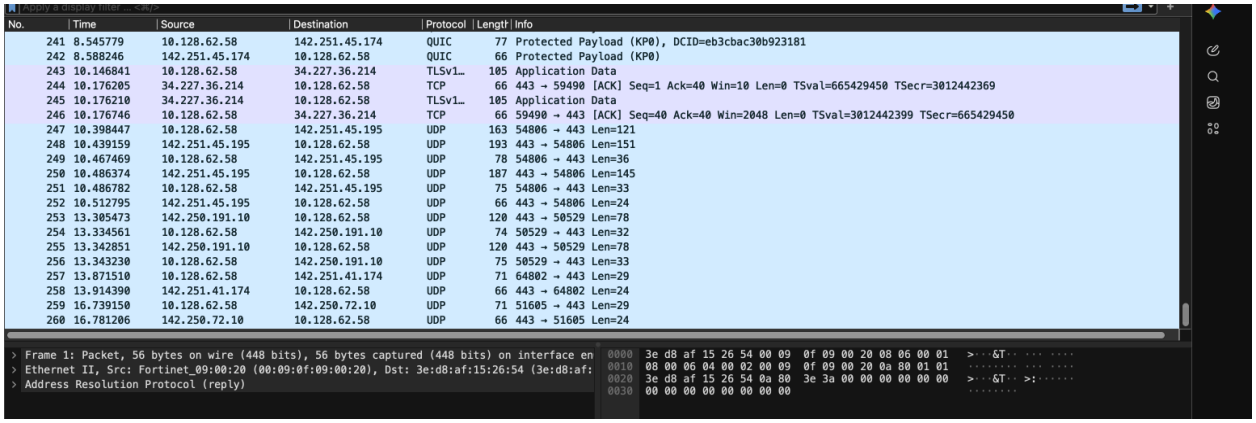
Screenshot: Compiler



Screenshot: Wireshark



Screenshot: Live capture



Issues:

There were a few issues that I encountered. The first was once I realized that my device had a software update that was necessary and I completed it. I still couldn't download the XCode Command line tools with the given command. I needed to go to StackOverflow to find a different command that basically did the same thing but accessed it in a different way that I don't understand. But it worked so I'll take it. I also had an issue with homebrew that was previously downloaded with some weird permissions for another class. With this I was having problems updating it and getting it to pass the tests. So my solution was to just delete all versions of it off of my device and reinstall it. I probably could have gone in and found the permissions to adjust but it was not worth the time.